[University Of Algeirs 1]

Department of Computer Science

MASTER 1 PROJECT REPORT

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Image Captioning Using Metadata-Filtered Retrieval-Augmented Generation (RAG)

Phase 1 — Dataset Exploration, Inspection, Preprocessing, Quality Assurance, and Multimodal Representation

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0.1

Abstract

Automatic image captioning is a fundamental task at the intersection of Computer Vision (CV) and Natural Language Processing (NLP). Its goal is to generate accurate, natural-language descriptions of visual content. Traditional deep learning models often struggle with uncommon objects or complex visual scenes because they rely entirely on parameters learned during training. Retrieval-Augmented Generation (RAG) addresses this challenge by retrieving relevant examples from an external knowledge base before generating a description.

This report presents **Phase 1** of a project titled **Image Captioning Using Metadata-Filtered Retrieval-Augmented Generation (RAG)**. Phase 1 focuses exclusively on building a clean, reliable, and well-structured multimodal foundation from the Microsoft Common Objects in Context (MS COCO) 2017 dataset. First, a persistent working environment was organized using Google Colab and Google Drive. Next, the 5,000 images of the `Val2017` dataset were analyzed based on object instances and unique object categories, classifying each image into Simple, Moderate, or Complex scene categories. Using an 80-column binary multi-label matrix and multi-label stratified sampling (`MultilabelStratifiedShuffleSplit`), a shared evaluation subset of 100 test images (40% Simple, 18% Moderate, and 42% Complex) was constructed and verified visually through a 10×10 contact sheet.

Furthermore, the `Train2017` dataset (118,287 images and 591,753 captions) was inspected and standardized to exactly five captions per image by sorting annotations deterministically by their official IDs, removing 318 excess captions to yield a pool of 591,435 captions. From this pool, 10,000 training images and their corresponding 50,000 captions were selected to form the retrieval knowledge base. All 50,000 captions underwent text normalization while preserving raw strings for traceability. Exploratory Data Analysis (EDA) on the 10,000 knowledge-base images revealed an average caption length of 10.46 words and an average object count of 7.26 objects per image. Finally, using the `clip-ViT-B-32` model via Sentence Transformers, 512-dimensional vector embeddings were generated for both the 10,000 images and 50,000 captions. The report also documents a methodological limitation regarding an overlap between a 100-image internal validation sample and the 10,000-image knowledge base, providing a clear remediation plan prior to Phase 2 retrieval implementation.

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Chapter 1

Introduction

1.0.1 General Background

Artificial intelligence has made major progress in processing visual data through Computer Vision (CV) and human language through Natural Language Processing (NLP). Combining these two domains is essential because human perception is naturally multimodal. Systems that jointly process images and text support important real-world applications, including assistive technologies for visually impaired users, content-based image indexing, and human-robot interaction.

1.0.2 Image Captioning

Image captioning is the task of automatically generating a natural-language sentence that describes the visual content of an image. Unlike object detection or simple image classification, image captioning requires two coordinated operations:

1. **Visual Understanding:** Identifying objects, their visual attributes, spatial relationships, and the overall scene context.
2. **Linguistic Generation:** Translating these visual observations into a grammatically correct sentence in human language.

1.0.3 Project Context

Standard encoder-decoder captioning models store all visual and linguistic knowledge inside their neural network weights. When presented with complex scenes, they often generate vague descriptions or hallucinate objects. **Retrieval-Augmented Generation (RAG)** addresses this limitation by searching an external knowledge base of existing images and captions to find relevant examples before generating a description. By adding **metadata filtering**—such as object categories or scene complexity—the retrieval step can select candidate captions that match both the visual appearance and the structural properties of the input image.

This project aims to develop an image captioning system based on a **Metadata-Filtered RAG** architecture using the MS COCO 2017 dataset. This report covers **Phase 1**, which establishes the dataset exploration, preprocessing, quality assurance, and multimodal representation pipeline.

1.0.4 Problem Statement

Preparing the MS COCO 2017 dataset for a Metadata-Filtered RAG pipeline requires solving five practical challenges:

- **Scale Constraints:** The full `Train2017` split (118,287 images and 591,753 captions) exceeds standard cloud notebook memory and storage limits.
- **Caption Count Inconsistencies:** While nominal specifications expect five captions per image, several images contain six or seven captions, which complicates indexing and biases retrieval probabilities.
- **Sampling Bias:** Selecting a 100-image test set via ordinary random sampling can under-represent rare object categories or skew scene complexity.
- **Textual Noise:** Raw captions contain inconsistent capitalization, punctuation, and spacing.
- **Cross-Modal Gap:** Raw images and text strings must be encoded into a shared numerical vector space without causing Out-Of-Memory (OOM) errors.

1.0.5 Project Objectives

The specific objectives of **Phase 1** are:

1. Configure a persistent workspace connecting Google Colab and Google Drive (`COCO_Project/`).
2. Classify all 5,000 `Val2017` images into Simple, Moderate, or Complex scene categories using object instance and category counts.
3. Construct a 5000×80 binary multi-label matrix mapping non-consecutive COCO category IDs to contiguous columns.
4. Select a representative 100-image test subset (500 reference captions) from `Val2017` using multi-label stratified sampling.
5. Standardize `Train2017` captions to exactly five per image via deterministic annotation ID sorting, and extract the 10,000-image (50,000-caption) knowledge base.
6. Visually audit the 100 test images using a 10×10 contact sheet and normalize all 50,000 training captions while preserving raw text.
7. Perform Exploratory Data Analysis (EDA) on the 10,000-image knowledge base and evaluate validation subset sampling.
8. Generate and verify 512-dimensional `clip-ViT-B-32` image and text embeddings.

1.0.6 Project Scope

This report covers Phase 1 exclusively. Phase 1 includes dataset inspection, directory setup, scene complexity analysis, multi-label matrix construction, stratified test set selection, caption standardization, visual quality assurance, text preprocessing, exploratory data analysis, and CLIP embedding generation. The RAG retrieval mechanism, metadata filtering engine, and caption generation model belong to future phases and have not yet been implemented.

1.0.7 Report Organization

Chapter 2 describes the MS COCO 2017 dataset, scene complexity rules, subset specifications, and data leakage prevention. **Chapter 3** details the eight implementation steps of Phase 1. **Chapter 4** presents the quantitative results, distributions, quality assurance checks, and methodological limitations. **Chapter 5** concludes the report and outlines future work.

Chapter 2

Dataset Description and Project Specifications

2.0.1 Introduction

In a Retrieval-Augmented Generation (RAG) system, the training dataset serves directly as the external knowledge base queried during inference. This chapter describes the MS COCO 2017 dataset splits, object annotations, scene complexity classification rules, project subset requirements, and data separation principles.

2.0.2 Overview of MS COCO 2017

The Microsoft Common Objects in Context (MS COCO) 2017 dataset captures everyday scenes containing multiple objects in natural contexts. Stored in JSON format, it provides two primary annotation types required for our project:

1. **Natural-Language Captions:** Human-written sentences describing the image content, used as the textual retrieval corpus.
2. **Object Instance Annotations:** Structured records identifying individual objects and their semantic categories, used for scene complexity analysis and metadata filtering.

2.0.3 Train2017 Dataset

The original `Train2017` caption annotation file contains **118,287 images** and **591,753 caption annotations**. In this project, `Train2017` serves as the source pool for the retrieval knowledge base. To fit within academic computing limits, a subset of **10,000 selected training images** and **50,000 captions** (exactly five captions per image) is used to form the knowledge base.

2.0.4 Val2017 Dataset

The original `Val2017` dataset contains **5,000 images**, each paired with object annotations and five reference captions (25,000 captions total). From `Val2017`, a shared subset of **100 images** (500 reference captions) was selected for testing across all project groups.

2.0.5 Object Annotations and Categories

Analyzing visual scenes requires distinguishing between **object instances** and **unique object categories**:

- **Object Instance:** A single annotated occurrence of an object in an image.
- **Unique Object Category:** The semantic class of the object (e.g., `person`, `car`, `dog`).

For example, an image showing four cars and two people contains 6 **object instances** ($N_{\text{obj}} = 6$) but only 2 **unique object categories** ($N_{\text{cat}} = 2$).

MS COCO defines exactly **80 object categories**. However, their official `category_id` integers in the JSON specification are **not consecutive** (they range from 1 to 90, with 10 integer IDs missing from early dataset revisions). Therefore, one must not confuse the 80 categories with the maximum ID value, and an explicit mapping between official category IDs and matrix column indices (0 to 79) is required.

2.0.6 Scene Complexity Classification

Each of the 5,000 `Val2017` images was assigned to one of three scene complexity classes based on its total object instances (N_{obj}) and unique object categories (N_{cat}), as defined in Table 2.1.

For **Simple** scenes, both conditions must hold simultaneously. For **Complex** scenes, satisfying either $N_{\text{obj}} \geq 8$ (crowded scene) or $N_{\text{cat}} \geq 4$ (high semantic variety) is sufficient. All remaining images fall into the **Moderate** class.

2.0.7 Caption Structure

Each image in MS COCO is associated with multiple independent human-written captions because the same visual scene can be described accurately in several ways. In the JSON files, each record stores the `image_id`, a unique official `annotation_id` (`id`), and the `caption` text. Although designed for five captions per image, raw `Train2017` contains a small number of images with six or seven captions, requiring standardization.

Table 2.1: Scene Complexity Classification Rules.

Class	**Total Objects ($N_{\mathbf{obj}}$)**	**Operator**	**Unique Categories ($N_{\mathbf{cat}}$)**	**Condition Requirement**
Simple	$N_{\text{obj}} \leq 3$	AND	$N_{\text{cat}} \leq 2$	Both conditions must be satisfied
Complex	$N_{\text{obj}} \geq 8$	OR	$N_{\text{cat}} \geq 4$	At least one condition must be satisfied
Moderate	All remaining combinations	–	–	Neither Simple nor Complex

2.0.8 Dataset Requirements

Table 2.2 summarizes the original MS COCO 2017 datasets alongside the required project subsets. In addition, the professor specified that the 100-image test subset must be shared across all groups and represent Simple, Moderate, and Complex scenes with diverse objects, interactions, and contexts.

Table 2.2: Original MS COCO 2017 Dataset vs. Selected Project Subsets.

Dataset / Subset	Source	Images	Captions/Img	Total Captions	Project Role
Original Train2017	`Train2017`	118,287	5 to 7	591,753	Raw training source pool
Standardized Pool	`Train2017`	118,287	Exactly 5	591,435	Standardized global pool
Knowledge Base	`Train2017`	10,000	Exactly 5	50,000	External RAG retrieval database
Original Val2017	`Val2017`	5,000	Exactly 5	25,000	Raw validation source pool
Selected Test Set	`Val2017`	100	Exactly 5	500	Shared evaluation benchmark

2.0.9 Data Separation and Leakage Considerations

Data leakage occurs when evaluation test samples appear in the training or retrieval database. In a RAG captioning system, if a test image were stored in the knowledge base, similarity search would retrieve its exact ground-truth captions, producing artificially inflated evaluation metrics. Constructing the 10,000-image knowledge base exclusively

from `Train2017` and selecting the 100 test images exclusively from the disjoint `Val2017` split prevents test data leakage.

2.0.10 Chapter Summary

This chapter defined the MS COCO 2017 partitions, distinguished object instances from the 80 non-consecutively numbered categories, specified the Simple/Moderate/Complex rules, and explained how separating `Train2017` and `Val2017` prevents test data leakage.

Chapter 3

Methodology and Implementation

3.0.1 Introduction

This chapter explains the eight technical steps implemented during Phase 1. For each operation, we explain what was done, why it was necessary, how it was implemented, what was obtained, and how it supports the overall RAG architecture.

3.0.2 Development Environment

The implementation was performed in **Google Colab** connected to **Google Drive**. Google Colab runs on temporary virtual machines whose local storage resets after inactivity, which would erase downloaded images and computed embeddings. Mounting persistent Google Drive storage preserved all files across sessions. After reconnecting Google Drive, the previously downloaded `Val2017` annotations were verified before execution continued.

3.0.3 Project Directory Organization

To prevent file overwrites and ensure reproducible paths across scripts, the root directory `COCO_Project/` on Google Drive was organized into eight dedicated folders: `raw/`, `images/train/`, `images/test/`, `annotations/`, `cleaned/`, `reports/`, `caption_selection/`, and `embeddings/`.

```
•••  Project folders created successfully in Google Drive!
     COCO_Project/
       reports/
       images/
         train/
         validation/
       annotations/
       cleaned/
       raw/
```

Figure 3.1: Folder structure of the MS COCO dataset used in the project.

3.0.4 Scene Complexity Analysis

All 5,000 `Val2017` images were analyzed using their object annotations in `instances_val2017.json`. For each image, two measurements were computed: (1) the total number of annotated object instances (N_{obj}) by counting annotation entries for that `image_id`, and (2) the number of unique object categories (N_{cat}) by counting distinct `category_id` values. Each image was then classified as **Simple** ($N_{\text{obj}} \leq 3$ and $N_{\text{cat}} \leq 2$), **Complex** ($N_{\text{obj}} \geq 8$ or $N_{\text{cat}} \geq 4$), or **Moderate** (all remaining images). This step provides objective metadata to ensure the test set spans diverse visual difficulty levels.

```
Complexity breakdown of the 5,000 val2017 images:
  - Simple: 1811
  - Moderate: 1064
  - Complex: 2077

✅ val2017 Matrix shape successfully built: (5000, 80)
```

Figure 3.2: Examples of simple, moderate, and complex scenes selected from the MS COCO dataset.

3.0.5 Multi-label Matrix Construction

Natural images are **multi-label** because multiple object categories can appear in the same photograph. To represent category presence across `Val2017`, a binary matrix $M \in \{0,1\}^{5000\times 80}$ was constructed with 5,000 rows (images) and 80 columns (COCO object categories). Because official COCO category IDs range up to 90 with gaps, the 80 valid category IDs were sorted and mapped to contiguous column indices $j \in \{0,\ldots,79\}$ via a lookup dictionary. Each matrix cell was populated as:

$$M_{i,j} = \begin{cases} 1, & \text{if category } j \text{ appears in image } i, \\ 0, & \text{otherwise.} \end{cases} \tag{3.1}$$

This binary representation allows stratification algorithms to balance all 80 object categories simultaneously when sampling a diverse test subset.

	image_id	person	bicycle	car	motorcycle	airplane	bus	train	truck	boat	...	toaster	sink	refrigerator			
0	391895	1	1	0	1	0	0	0	0	0	...	0	0	0	0	0	0
1	522418	1	0	0	0	0	0	0	0	0	...	0	1	0	0	0	0
2	184613	1	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0
3	318219	1	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0
4	554625	1	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0
5	574769	1	0	0	0	0	0	0	0	0	...	0	1	1	0	1	0
6	60623	1	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0
7	309022	0	0	0	0	0	0	0	0	0	...	0	1	0	0	0	0
8	5802	1	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0
9	222564	1	0	0	0	0	0	0	0	0	...	0	0	0	0	0	0

10 rows × 81 columns

Table successfully saved to: /content/drive/MyDrive/COCO Project/reports/matrix preview.csv

Figure 3.3: Sample of the multi-label matrix showing the binary representation of object categories in COCO images.

3.0.6 Test Subset Selection

Selecting 100 images from 5,000 via ordinary random sampling can omit rare categories or yield an unbalanced subset. Instead, **multi-label stratified sampling** was performed using `MultilabelStratifiedShuffleSplit` from the `iterative-stratification` Python library, guided by the 5000×80 matrix. Because each image contains a bundle of co-occurring labels, multi-label stratification preserves category prevalence **approximately** rather than exactly, and initially produced a candidate set of **106 images**.

To meet the exact requirement of 100 images, **6 images were removed using a fixed random seed of 42**. A fixed seed guarantees exact reproducibility across runs, although removing 6 images after stratification slightly modifies the final category distribution. The final 100-image test subset contained **40 Simple (40%)**, **18 Moderate (18%)**, and **42 Complex (42%)** scenes (500 reference captions).

3.0.7 Caption Standardization

Inspection of the 118,287 `Train2017` images showed that 117,972 images had 5 captions, 312 had 6 captions, and 3 had 7 captions (591,753 total). Having exactly five captions per image provides a consistent data structure and simplifies indexing and comparison across images.

To standardize the dataset deterministically, each image's captions were sorted in ascending order by their official `annotation_id`, and the first five captions were retained while $312 + (3 \times 2) = 318$ excess captions were excluded. Note that `annotation_id` is an administrative identifier used strictly for reproducible ordering and must **not** be interpreted as a quality ranking. This produced a **standardized global annotation pool of 591,435 captions** $(118,287 \times 5)$, from which the **final knowledge base of 50,000 captions for 10,000 selected images** was extracted.

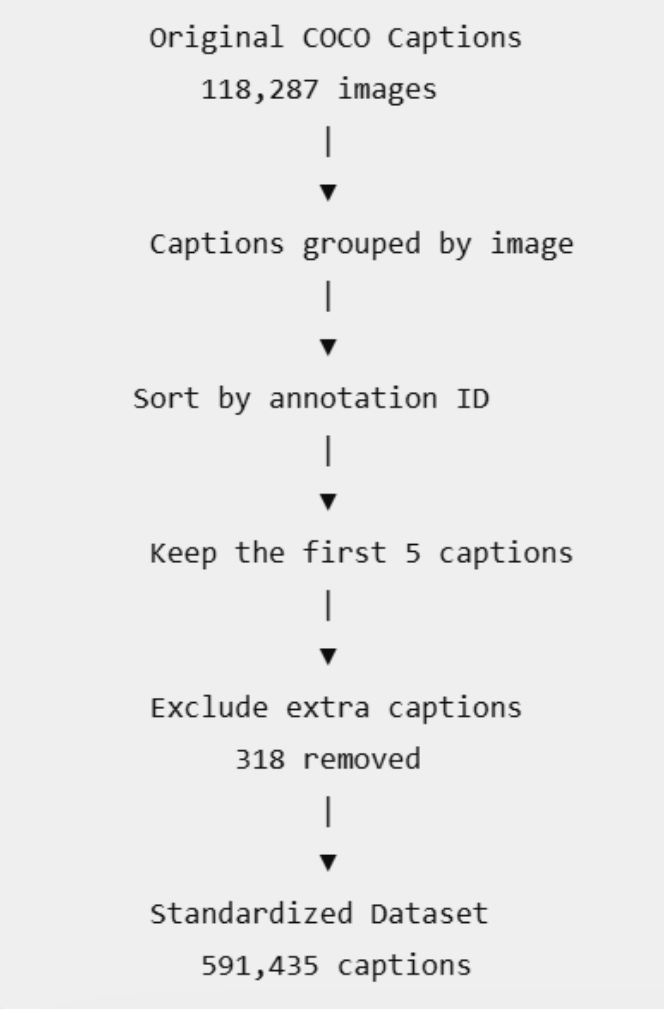


Figure 3.4: Caption standardization workflow showing how exactly five captions were retained for each training image.

3.0.8 Visual Quality Assurance

The 100 selected `Val2017` test images were downloaded and arranged into a 10×10 **contact sheet** for human visual inspection. Visual inspection after quantitative sampling is necessary to verify that image files are not corrupted and to confirm visually that the subset includes diverse scene contexts, simple and complex layouts, multiple objects, and object interactions. While visual inspection provides practical quality assurance, it does

not mathematically prove perfect dataset quality.

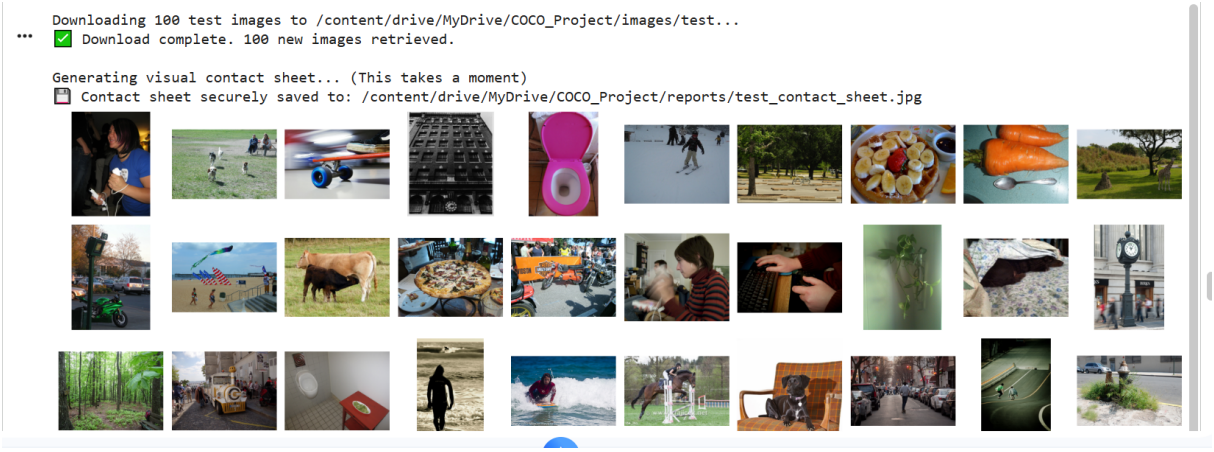


Figure 3.5: Contact sheet showing the 100 selected MS COCO Val2017 test images.

3.0.9 Text Preprocessing

All 50,000 knowledge-base captions underwent three normalization steps to remove superficial formatting variations (note that text preprocessing standardizes text representation but does not by itself prevent data leakage):

1. **Lowercasing:** Converts uppercase letters to lowercase.
2. **Punctuation Removal:** Strips punctuation characters such as `!`, `.`, and `,`.
3. **Whitespace Normalization:** Trims leading/trailing spaces and collapses multiple spaces.

For example, the raw caption `"A Man Is Riding A Bike!"` is transformed into the cleaned caption `"a man is riding a bike"`. Both `raw_captions` and `clean_captions` were preserved to ensure data traceability, preserve original grammatical formatting, allow direct side-by-side comparison, and guarantee reproducibility.

> **[PLACEHOLDER — Figure 7: Text Preprocessing Example]**
> Comparison of a raw caption (`"A Man Is Riding A Bike!"`) and its cleaned version (`"a man is riding a bike"`).

Figure 3.6: Text Preprocessing Example comparing `raw_captions` and `clean_captions`.

3.0.10 Validation Subset and Exploratory Data Analysis

A 100-image validation subset was sampled from the 10,000 knowledge-base image IDs using a fixed random seed of 42 via `validation_100_ids = random.sample(train_10000_ids, 100)`. Because these 100 images were **not removed** from the 10,000-image knowledge

base, the validation subset overlaps completely with the knowledge base. This differs from the professor's requirement that the 100 `Train2017` validation images be disjoint from the knowledge base. Using these overlapping images for validation while their information remains in the retrieval database creates a data leakage risk; a corrected implementation must sample validation IDs from `Train2017` images outside the 10,000 knowledge-base IDs.

Next, Exploratory Data Analysis (EDA) was conducted on the 10,000 knowledge-base images and 50,000 cleaned captions, measuring an **average caption length of 10.46 words** (useful for understanding language conciseness and encoder sequence limits) and an **average object count of 7.26 objects per image** (useful for quantifying visual complexity). Because an arithmetic average only describes the central value and hides spread or skewness, two histograms were also generated to show the full distributions.

3.0.11 CLIP Embedding Generation

An **embedding** is a numerical vector that represents the semantic meaning of an image or sentence as coordinates in a mathematical space. Using the **Sentence Transformers** library, we loaded the `clip-ViT-B-32` model (based on the CLIP architecture). CLIP trains an image encoder and a text encoder jointly to project both images and text into a **shared 512-dimensional embedding space**, where each of the 512 dimensions captures latent visual-semantic features. In this shared space, semantically similar images and captions lie close together and can be compared directly via cosine similarity:

The cosine similarity between two embeddings can be calculated by dividing their dot product by the product of their vector lengths.

Embeddings were generated for all 50,000 cleaned captions and 10,000 knowledge-base images. Image encoding was performed in **batches of 500** to limit RAM/VRAM usage and avoid Google Colab Out-Of-Memory errors. Both output files—`train_text_embeddings.npy` with shape `(50000, 512)` and `train_image_embeddings.npy` with shape `(10000, 512)`—were verified to ensure no missing rows or corrupted dimensions.

```
Loading image 574769...
Generating mathematical embeddings...

--- Embedding Verification ---
Image vector shape: (512,) (Should be 512)
Text matrix shape: (3, 512) (Should be 3 x 512)
First 5 dimensions of the image vector:
[-0.522  0.371 -0.225  0.405 -0.184]
```

Figure 3.7: Verification of CLIP image and text embeddings, showing their 512-dimensional vector representations.

3.0.12 Chapter Summary

This chapter detailed the persistent environment setup, scene complexity classification, 5000×80 multi-label matrix construction, stratified test sampling, caption standardization, 10×10 visual quality assurance, text preprocessing, EDA and validation overlap analysis, and batch-processed `clip-ViT-B-32` embedding generation.

Chapter 4

Results and Quality Assurance

4.0.1 Introduction

This chapter presents the quantitative results, distributions, and quality assurance verifications obtained in Phase 1, explaining both what the findings demonstrate and their boundaries.

4.0.2 Dataset Inspection Results

Inspection of the raw `Train2017` caption file confirmed 118,287 images and 591,753 captions. As shown in Table 4.1, 315 images contained more than five captions, demonstrating why raw annotations cannot be indexed with a fixed stride of 5 without prior standardization.

Table 4.1: Caption Count Distribution in Original `Train2017` Dataset.

Caption Count per Image	Number of Images	Percentage of Images	Total Captions
Exactly 5 captions	117,972	99.7337%	589,860
Exactly 6 captions	312	0.2638%	1,872
Exactly 7 captions	3	0.0025%	21
Total	**118,287**	**100.0000%**	**591,753**

4.0.3 Caption Standardization Results

Sorting captions by official `annotation_id` and retaining the first five captions per image excluded $312 + (3 \times 2) = 318$ captions, producing a standardized global pool of $118,287 \times 5 = 591,435$ captions. From this pool, 10,000 images and 50,000 standardized captions were extracted for the knowledge base. This demonstrates structural consistency across all images, though it does not imply that the 318 excluded captions were of lower linguistic quality.

4.0.4 Test Subset Distribution

After multi-label stratification on the 5000×80 `Val2017` matrix yielded 106 candidates and 6 images were removed (seed = 42), the final 100-image test subset distribution shown in Table 4.2 was obtained.

Table 4.2: Final Test Subset Distribution Across Scene Complexity Classes.

Scene Category	Classification Rule Summary	Number of Images	Percentage	Refere
Simple	$N_{\text{obj}} \leq 3$ AND $N_{\text{cat}} \leq 2$	40	40%	
Moderate	Remaining images	18	18%	
Complex	$N_{\text{obj}} \geq 8$ OR $N_{\text{cat}} \geq 4$	42	42%	
Total	**All Classes**	**100**	**100%**	

Table 4.2 demonstrates strong representation of both Simple (40%) and Complex (42%) scenes alongside Moderate (18%) scenes, ensuring rigorous evaluation across complexity tiers, while acknowledging that multi-label stratification and post-split trimming preserve label prevalence approximately rather than exactly.

4.0.5 Visual Inspection Results

Visual inspection of the 10×10 test contact sheet (Figure **??**) confirmed that all 100 downloaded images were intact and represented diverse indoor/outdoor environments, varied object categories, simple single-subject photos, and complex multi-object interactions.

4.0.6 Text Preprocessing Results

Processing the 50,000 knowledge-base captions produced 50,000 normalized strings in `clean_captions` paired 1-to-1 with `raw_captions` in `COCO_Project/cleaned/`, while preserving a one-to-one correspondence between each original caption and its normalized version.

4.0.7 Exploratory Data Analysis

Table 4.3 summarizes the EDA measurements across the 10,000 knowledge-base images and 50,000 cleaned captions, complemented by the histograms in Figure **??** and Figure **??**.

Table 4.3: Exploratory Data Analysis (EDA) Summary Statistics on the Knowledge Base.

EDA Metric	Evaluated Scope	Sample Size	Measured Average	Visua
Average Caption Length	Cleaned KB Captions	50,000 captions	**10.46 words**	Histog (Fig- ure ??
Average Object Count	Selected KB Images	10,000 images	**7.26 objects/img**	Histog (Fig- ure ??

```
- 1. Creating Validation Split ---
| Extracted 100 validation IDs from training set.
| Saved to: /content/drive/MyDrive/COCO_Project/reports

- 2. Analyzing Caption Lengths ---
tal captions analyzed: 50,000
erage caption length: 10.46 words
| Histogram saved to: /content/drive/MyDrive/COCO_Proje
```

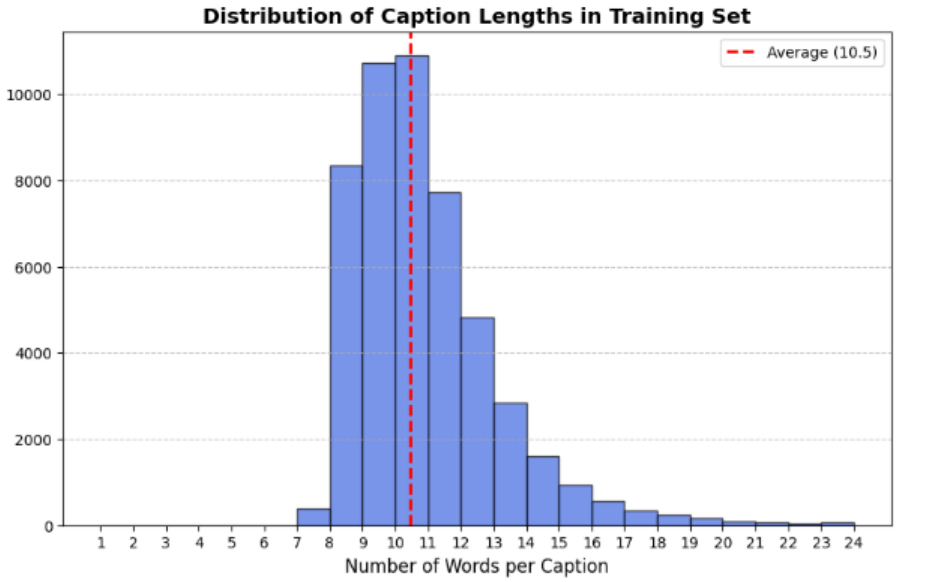


Figure 4.1: Distribution of caption lengths across the 50,000 training captions after text preprocessing.

```
Loading data for object count analysis...
Total images analyzed: 10,000
Average objects per image: 7.26
📊 Histogram securely saved to: /content/drive/MyDrive
```

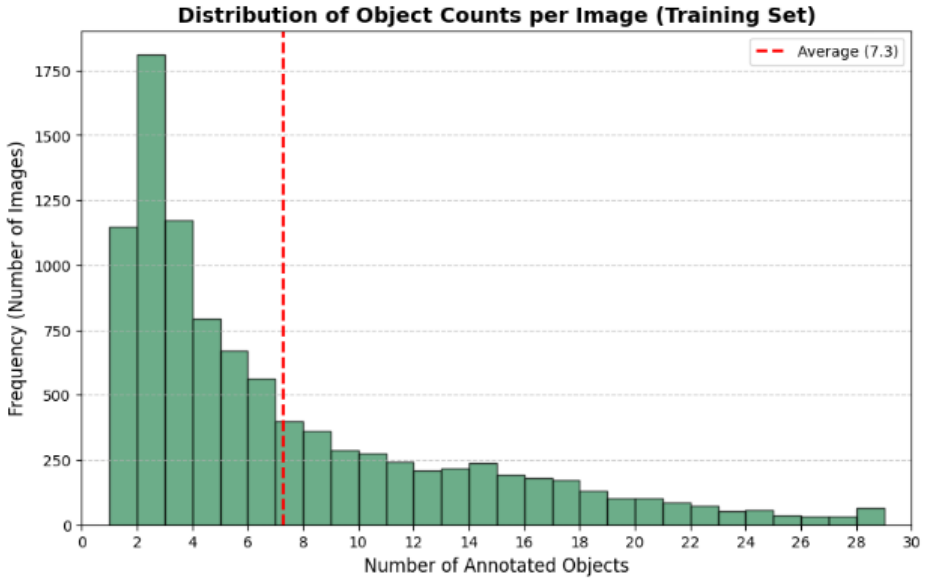


Figure 4.2: Distribution of annotated object counts across the 10,000 selected training images.

These results demonstrate that reference captions are concise single sentences (10.46 words on average) and that the knowledge base contains visually dense scenes (7.26 objects per image on average).

4.0.8 CLIP Embedding Results

Table 4.4 reports the verified shapes of the generated `clip-ViT-B-32` embeddings. Confirming exact dimensions of `(50000, 512)` and `(10000, 512)` demonstrates complete encoding of the knowledge base without missing rows, though downstream retrieval effectiveness remains to be evaluated in Phase 2.

Table 4.4: Generated CLIP Embedding Specifications and File Names.

Embedding Type	Model ID	Library	Batch Size	File Name
Text Embeddings	`clip-ViT-B-32`	Sentence Transformers	Standard	`train_text_embedd`
Image Embeddings	`clip-ViT-B-32`	Sentence Transformers	500	`train_image_embed`

4.0.9 Quality Assurance and Verification

Table 4.5 summarizes all Phase 1 tasks and their verification status.

Table 4.5: Summary of Completed Phase 1 Tasks and Verification Status.

Step	Phase 1 Task	Verification Method	Confirmed Outcome	Status
1	Environment & Storage	Path checks on Google Drive (`COCO_Project/`)	8 folders & `Val2017` JSON verified	Completed
2	Complexity & Matrix	Category mapping & matrix dimension check	5000 × 80 binary matrix built	Completed
3	Test Set Selection	Stratified split & seed-42 trimming check	100 `Val2017` images (40/18/42%)	Completed
4	Caption Standardization	Count audit after `annotation_id` sort	318 removed; 591,435 pool; 50,000 KB	Completed
5	Visual QA	Visual audit of 10 × 10 contact sheet	100 test images intact & diverse	Completed
6	Text Preprocessing	Comparison of `raw` and `clean` captions	50,000 normalized; raw preserved	Completed
7	Validation & EDA	Statistics, histograms, and ID audit	10.46 words; 7.26 objects; Overlap noted	Completed
8	CLIP Embeddings	Batch execution (500) & `.shape` check	`(50000, 512)` & `(10000, 512)` verified	Completed

4.0.10 Limitations and Challenges

1. **Validation / Knowledge-Base Overlap:** Sampling the 100 internal validation IDs directly from `train_10000_ids` without removal means the validation subset overlaps with the knowledge base, creating a data leakage risk during validation tuning (whereas the 100-image `Val2017` test set remains completely independent).
2. **Approximate Multi-Label Stratification:** Multi-label stratification preserves category distributions approximately, and removing 6 images from 106 candidates slightly alters the final proportions.
3. **Pending Retrieval Evaluation:** Phase 1 stops at verified embeddings; end-to-end RAG retrieval and caption generation have not yet been evaluated.

4.0.11 Chapter Summary

This chapter presented the verified results of Phase 1 across caption standardization, test set stratification, visual inspection, text normalization, EDA, and CLIP embedding generation, along with an honest assessment of the validation overlap limitation.

Chapter 5

Conclusion and Future Work

5.0.1 Conclusion

Phase 1 established a clean, reproducible multimodal data foundation from MS COCO 2017 for our Metadata-Filtered RAG image captioning system. By combining scene complexity rules, an 80-category binary matrix, multi-label stratified sampling, deterministic caption standardization, visual and textual quality assurance, exploratory data analysis, and batch-processed `clip-ViT-B-32` embeddings, the dataset is fully prepared for similarity search and retrieval modeling.

5.0.2 Summary of Completed Work

We organized a persistent 8-folder Google Drive workspace, classified all 5,000 `Val2017` images by scene complexity, built a 5000×80 multi-label matrix, selected and visually verified the 100-image `Val2017` test set (40% Simple, 18% Moderate, 42% Complex), standardized `Train2017` to 591,435 captions (extracting 10,000 images and 50,000 captions for the knowledge base), normalized all 50,000 captions while preserving raw text, measured EDA distributions (10.46 words/caption and 7.26 objects/image), and generated verified `(50000, 512)` text and `(10000, 512)` image embeddings.

5.0.3 Main Contributions

Phase 1 delivers: (1) a standardized 10,000-image / 50,000-caption knowledge base encoded in a 512-dimensional CLIP vector space, (2) a 100-image test subset selected from the separate `Val2017` split and organized by scene complexity, and (3) reusable scene complexity and category mapping rules for future metadata filtering.

5.0.4 Current Limitations

The primary limitation is the overlap between the 100-image internal validation subset and the 10,000-image knowledge base, alongside the approximate nature of post-trimmed multi-label stratification and the fact that retrieval and generation pipelines belong to Phase 2.

5.0.5 Future Work

Future work in Phase 2 will:

1. **Correct the Validation Overlap:** Re-sample the 100 validation images from `Train2017` IDs outside the 10,000 knowledge-base images.
2. **Organize Knowledge-Base Metadata:** Structure object category and scene complexity metadata for all 10,000 training images.
3. **Build the Retrieval Mechanism:** Implement cosine similarity search over the `(10000, 512)` image and `(50000, 512)` text embeddings.
4. **Implement Metadata-Filtered RAG:** Combine vector similarity retrieval with metadata filtering and integrate a language generation model.
5. **Evaluate Caption Quality:** Benchmark generated captions on the 100 `Val2017` test images across different system configurations using standard captioning metrics.

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